

Rohan Mehra

Graduate Student in Computer Science

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Education

- 2025 – present **ISIMA, Université Clermont Auvergne** (Ranked 8/34)
MS in Computer Science
- 2021 – 2025 **Indian Institute of Science Education and Research (IISER) Bhopal** (Grade: 8.92/10)
BS in Electrical Engineering and Computer Science

Research Experience

- Mar. – present **LIFAT, CNRS, Université de Tours, M2 Intern**
- 2026
- Advised by Prof. Gilles Venturini as part of RFAI team.
 - Working on AI-assisted image recognition of bumblebees (*bombus*).
 - Using transfer learning, and explainable AI approaches.
- 2025 – 2026 **Institut Pascal, CNRS, Université Clermont Auvergne, M2 Project Student / Intern**
- & May – July
- 2024
- Advised by Dr. Mathieu Labussière as part of ComSee team.
 - Worked on deep learning models on SWIR images' for road perception in adverse weather conditions.
 - 📄 ○ Models worked on include Grounded SAM, YOLO, RF-DETR [Sem. Segmentation] and CUT, CycleGAN, Pix2Pix, BBDM [GANs & diffusion-based]
- Jan. – April **IISER Bhopal, BS Project Student**
- 2025
- Advised by Dr. Arpit Sharma at the Dept. of EECS.
 - 🔗 📄 ○ Worked on using Zero-Shot Learning model for Sector Classification, utilizing the CrowdRE dataset.
 - Used models including Sentence-BERT, All-MiniLM-L12, Bert4RE and BERTOverflow
- May – July **Indian Institute of Space Science and Technology, Intern**
- 2023
- 📄 ○ Advised by Prof. Deepak Mishra as part of the CVVR Lab
 - 📄 ○ Worked on robotics and implementation of SLAM through Kalman filter Methods.
 - Used Robot Operating System (ROS), RViz, and Gazebo.

Selected Publications

Full list available on [Google Scholar](#)

- 🔗 **Rohan Mehra**, Alexandre Riffard, Yannis Loumouamou, Mathieu Labussière, "Analysis of SWIR Imaging Detection Performance Under Adverse Environmental Conditions for Autonomous Driving Systems", **Under review at IEEE Transactions on Intelligent Vehicles (T-IV)**, 2026.
- 🔗 Yannis Loumouamou, Alexandre Riffard, **Rohan Mehra**, Mathieu Labussière, "Experimental study of YOLOv8 fine-tuning for RGB-SWIR multimodal road detection", *Reconnaissance des Formes, Image, Apprentissage et Perception (RFIAP)*, 2026.
- 🔗 **Rohan Mehra**, Alexandre Riffard, Mathieu Labussière, Pierre Duthon, Romuald Aufrère, "Would SWIR modality help for detection and segmentation in harsh weather conditions? An experimental study.", *Proceedings of the IEEE/CVF International Conference on Computer Vision Workshops (ICCVW)*, 2025. [Oral]

Teaching Experience

- 2024 & 2023 **Basic Electronics, IISER Bhopal** (× 2)
- Teaching Assistant for an undergraduate class of 180 students for 2024-25/23-24 Semester I.
 - Conducted tutorial sessions, prepared assignment solutions, and provided one-to-one guidance.

Minor Course Projects

- 2025 **Risk-Aware Multi-Epoch Package Delivery Planning** *Optimization @ IISER-B*
🔗 📄
 - Used greedy algorithm to solve package delivery problems while balancing risk and reward.
 - Developed a GUI-based visualization tool using Tkinter and Matplotlib.
- 2024 **Training Multi-Robot Warehouse (RWARE) Environment** *RL @ IISER-B*
🔗
 - Trained robots for optimized coordination in the RWARE environment.
 - Employed Deep Q-Network (DQN) for efficient learning.
- 2024 **Implementation of Grover's Algorithm** *Quantum Computing @ IISER-B*
📄
 - Implemented Grover's Algorithm for the 3-SAT problem demonstrating quantum speedup.
 - Utilized PennyLane for designing quantum circuits.
- 2023 **Prediction of Mortality Rate of Heart Failure Patients in ICU** *ML @ IISER-B*
🔗
 - Built a classification model to predict ICU mortality in heart failure patients.
 - Utilized Random Forest, Logistic Regression, and AdaBoost etc. for model development.

Technical Skills

Languages Python (with PyTorch framework), C++, C, Java
Software ROS, Gazebo, MATLAB, Git, $\text{\LaTeX}$, Wolfram Mathematica

Events

- Mar. 2026 **ELLIS Winter School on Foundation Models (FoMo)**
Selected to attend the FoMo Winter School hosted by the ELLIS unit Amsterdam, among a global pool of applicants (~250 participants).
- Feb. 2026 **Researchers' Day – DATA Program I-SITE CAP 20-25**
Presented a poster at PhD day, Clermont-Ferrand (sole master student).

Fellowships / Achievements

- 2025 **MIAI Cluster Excellence Scholarship**
Selected for Excellence Scholarship program offered by the Multidisciplinary Institute in Artificial Intelligence (MIAI) in France to pursue master's program, awarded to ~20 candidates university wide.
- 2024 **IASc-INSA-NASI Summer Research Fellowship** *[Declined]*
Selected for summer research fellowship jointly offered by the Indian Academy of Sciences, Indian National Science Academy, and National Academy of Sciences, awarded to the top ~5% of applicants.
- 2021 **Indian Maritime University Undergraduate Admissions**
Secured All India Rank 9 out of 14,108 ranked candidates, placing in the top 0.1% nationwide.

Positions of Responsibility / Extra Curriculars

- 2023 – 2024 **Cultural Secretary, IISER Bhopal**
Elected representative of the student body for organizing all cultural events at IISER Bhopal. Led a team of 70 performers to the the Inter IISER Cultural Meet 2023 held at IISER Mohali.
- 2022 **NCC 'B' Certificate, National Cadet Corps, India**
Completed 'B' division NCC military training from 8 UP BN NCC, Aligarh.

Other

Languages English (fluent, TOEFL 99/120), Hindi (Native), French (Basic)
Hobbies Table Tennis, Quizzing